

Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}\bar{W}_k(\sigma^+, \sigma^-) &= \left[\frac{1}{U_k(\sigma^+) + v_k} + \frac{1}{U_k(\sigma^-) + v_k} - \frac{1}{u_{k-1} + v_k} \right]^{-1} - v_k \\ U_{k+1}(z) &= \left[\frac{1}{U_k(z) + v_k} + \frac{1}{u_k + v_k} - \frac{1}{u_{k-1} + v_k} \right]^{-1} - v_k \\ u_{k+1} &= \left[\frac{2}{u_k + v_k} - \frac{1}{u_k + v_{k-1}} \right]^{-1} - v_k\end{aligned}$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}D &= \sum_{x,y,z \in \mathbb{F}_q^\times} \theta(x^d z) \theta(ax^{d-1} z) \theta(bz) \theta(-zy^2) \\ &= \frac{1}{(q-1)^4} \sum_{l,m,n,k=0}^{q-2} G_{-m} G_{-l} G_{-n} G_{-k} T^l(a) T^n(b) T^k(-1) \\ &\times \sum_{x \in \mathbb{F}_q^\times} T^{l(d-1)+md}(x) \sum_{y \in \mathbb{F}_q^\times} T^{2k}(y) \sum_{z \in \mathbb{F}_q^\times} T^{l+m+n+k}(z).\end{aligned}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}\frac{\delta \hat{A}}{\delta \tau} &= \rho \varepsilon_\rho - \varepsilon - \lambda \equiv (p - \lambda) = 0, \quad \frac{\delta \hat{A}}{\delta \lambda} = (x_m - \tau) = 0, \\ \frac{\delta \hat{A}}{\delta u} &= (u - \nu) = 0, \quad \frac{\delta \hat{A}}{\delta \nu} = (x_t - u) = 0, \\ \frac{\delta \hat{A}}{\delta S} &= -(r_t + T) = 0, \quad \frac{\delta \hat{A}}{\delta r} = S_t = 0.\end{aligned}$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\sum_{(d^\bullet; d^\infty)} \sum_{(d^1, \bullet, \dots, d^n, \bullet; d^\infty, \bullet)} (-1)^{\sum_{i=1}^n (l_i - 1) + l_\infty} \left(\prod_{i=1}^{\infty} \prod_{j=1}^{l_i} 1_{d^{i,j}} \right) \star 1_{d^\infty, \infty}^\sigma.$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned}
& c_{1a}f_{-}(u_{+})\mathbf{U}_{a-1}\mathbf{U}_{a-1}' + c_{2a}f_{+}(u_{-})\mathbf{U}_{a-1}\mathbf{D}_{a+1}' + f_{-}\mathbf{X}_{a,a-2}^{a-1}\mathbf{X}_{a-2,a}'^{a-1} \\
& = c_{2a}f_{-}(u_{+})\mathbf{D}_{a+1}'\mathbf{D}_{a+1} + c_{1a}f_{+}(u_{-})\mathbf{U}_{a-1}'\mathbf{D}_{a+1} + f_{-}\mathbf{X}_{a,a+2}'^{a+1}\mathbf{X}_{a+2,a}^{a+1}
\end{aligned}$$